



Transformer-variational autoencoder for estimating individual treatment effect using causal inference framework

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Abstract

Causal inference serves as a critical framework in diverse domains, including healthcare, economics, and the social sciences. Within this context, data-driven machine learning models have gained prominence for estimating individual treatment effects (ITEs). Recently, the Causal Effect Variational Autoencoder (CEVAE) was proposed to infer ITEs from observational data with unobserved confounders. However, CEVAE faces limitations in certain scenarios for instance, it struggles to accurately infer latent confounders in high-dimensional data, leading to biased causal effect estimates. To address these challenges, we introduce the Transformer Causal Effect Variational Autoencoder (TCE-VAE), a novel architecture that integrates transformer-based encoder-decoders with variational autoencoders. The core innovation of TCE-VAE lies in its use of a self-attention mechanism to directly estimate causal effects by capturing complex dependencies and interactions within the data. This approach enhances robustness in learning latent confounders, particularly in high-dimensional settings. We evaluate TCE-VAE on the Infant Health and Development Program (IHDP) dataset. Experimental results demonstrate that the proposed approach outperforms popular state-of-the-art models in terms of average treatment effect estimation. Furthermore, our method exhibits superior precision and robustness in individual treatment effect estimation compared to existing approaches.

Keywords Causal inference · Confounding variables · Deep learning · Self-attention mechanism · Transformer · Variational autoencoder

1 Introduction

Causal inference enables researchers to rigorously identify cause-and-effect relationships between variables. In medical research, this framework is pivotal for evaluating how interventions such as novel therapies impact patient outcomes. A standard approach involves randomly assigning participants to a treatment group (exposed to the intervention) or a control group (maintained under baseline conditions). As illustrated in Fig. 1, the treatment variable (t) represents the intervention, while the outcome variable (y) corresponds to the measured health parameter (e.g., blood pressure). Latent

confounding variables (Z), which are unobserved but influence both t and y , along with their observable proxies (X), are explicitly modeled to account for potential biases.

Through rigorous statistical adjustment for confounders and comparative analysis of outcomes across treatment and control groups, researchers can quantify the true causal effect of an intervention. A practical illustration of this methodology involves evaluating changes in blood pressure clinically relevant endpoint, before and after administering treatment. These analyses not only generate robust, evidence-based insights into therapeutic efficacy but also directly inform healthcare strategies and policy formulation, ensuring decisions are grounded in causal validity rather than mere correlation.

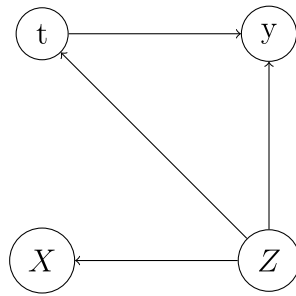
A fundamental challenge in causal inference lies in disentangling true causal effects from mere statistical associations. The gold standard for achieving this is Randomized Controlled Trials (RCTs) [1], where participants in a selected cohort are randomly assigned to treatment or control groups. The causal effect is then inferred from the observed outcome

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Fig. 1 Causal inference with treatment variable (t), outcome (y), latent or hidden confounding variable (Z) and (X) is proxy views on hidden confounding variable (Z)



difference between these groups. However, in observational studies, methods such as Inverse Probability of Treatment Weighting (IPTW) and Doubly Robust Estimators (e.g., Augmented IPW) are widely employed to adjust for confounding variables [2–4]. A persistent challenge in applying these techniques is the correct specification of propensity score and outcome models. Misspecification—common in high-dimensional or non-linear settings, can lead to biased estimates and erroneous conclusions [5, 6]. The complexity of real-world data, characterized by high dimensionality and intricate causal relationships, exacerbates these challenges. This underscores the need for advanced causal inference methodologies that are both robust to model misspecification and capable of capturing complex interactions in modern applications.

In recent years, the integration of machine learning (ML) into causal inference has opened new avenues for addressing complex causal relationships in high-dimensional settings. Pioneering this effort, [7] introduced causal trees and forests, leveraging ensemble random forest techniques to estimate heterogeneous treatment effects with robust statistical inference. Expanding on this foundation, [8] developed generalized random forests—building on the seminal work of [9] and later advancements by [10]—to extend forest-based methods to a broader class of causal parameters, such as conditional average treatment effects. These approaches have demonstrated efficacy in high-dimensional environments with diverse confounding structures and heterogeneous treatment effects. Concurrently, [11] proposed the double machine learning framework, which combines orthogonalization techniques with flexible ML models. This method enables valid inference on treatment effects even when both the propensity score and outcome models are high-dimensional or nonparametric.

Deep learning (DL) has emerged as a transformative force in causal inference, offering powerful tools to model complex, nonlinear relationships. A seminal advancement in this domain is the work of [12], who proposed neural network architectures to learn balanced representations of covariates, effectively addressing the challenge of unobserved counterfactuals in estimating individual treatment effects (ITEs). Subsequent innovations have extended DL

to specialized data structures: for instance, graph neural networks (GNNs) have been adapted by [13] to model spillover effects and interdependencies in networked data, enabling precise estimation of treatment effects in settings with social or spatial interactions. Recent years have seen the rise of transformer architectures in causal inference. [14] introduced the Causal Transformer, which leverages self-attention mechanisms to model long-range dependencies in longitudinal data, improving counterfactual outcome prediction over time. Further advancing this paradigm, [15] developed TransTEE, a transformer-based framework for heterogeneous treatment effect (HTE) estimation that accommodates multiple treatment types. Concurrently, [16] proposed Causal Inference with Attention (CInA), a meta-learning approach that enables zero-shot causal inference on novel tasks by leveraging attention-based task representations. These DL-driven methodologies collectively address longstanding barriers in causal inference—such as high-dimensionality, temporal dependencies, and heterogeneous effects—while maintaining robustness to complex confounding structures.

Generative models, particularly variational autoencoders (VAEs) and generative adversarial networks (GANs), have emerged as promising tools for causal effect estimation. A VAE’s architecture—an encoder, decoder, and latent space, enables the identification of proxy variables to approximate unobserved confounders. While this approach mitigates bias from hidden confounding, residual biases often persist, necessitating complementary methods for robust causal inference. For instance, [17] introduced the Causal Effect VAE (CEVAE), which leverages latent variable modeling to adjust for confounders. Similarly, proximal causal inference frameworks ([18]; [19]; [20]) formalize the use of proxy variables to reduce confounding bias in observational studies. The Balancing Covariates Automatically using Supervision (BCAUS) was discussed by [21], using deep-neural-network-based propensity models. Balancing Covariates Automatically Using Self-Supervision (BCAUSS), extended a multi-task deep neural network for improved handling of dissimilarity and positivity assumption violations discussed by [22].

A critical challenge in VAE-based causal inference lies in balancing covariate distributions between treatment and control groups. Recent advancements aim to refine covariate balancing techniques, enhancing the validity of causal effect estimates. Despite their strengths, traditional machine learning methods—including neural networks (NNs), GANs, and VAEs—face limitations in capturing long-range dependencies and structural causal relationships, particularly when latent confounders induce spurious correlations between treatment and outcomes.

In this study, we propose a novel methodology that integrates the transformer self-attention mechanism into a variational autoencoder (VAE) framework to address challenges in modeling long-range dependencies and latent confounding. The proposed TCE-VAE approach leverages self-attention to dynamically identify and adjust for latent variables critical for capturing unobserved confounders that influence both treatment assignment and outcomes. By explicitly modeling these hidden factors, TCE-VAE reduces bias and enhances the robustness of causal effect estimates, even in high-dimensional and structurally complex datasets. The self-attention mechanism enables the model to prioritize influential data patterns while suppressing spurious correlations, thereby improving the validity of causal inferences. Evaluated on the Infant Health and Development Program (IHDP) benchmark dataset, TCE-VAE demonstrates superior performance, yielding less biased and more precise estimates compared to state-of-the-art baselines. This advancement highlights the potential of transformer-enhanced architectures to overcome persistent limitations in causal inference, particularly in settings with unmeasured confounding and intricate dependency structures.

Three major contributions of this work are summarized as follows:

- **Integration of Transformers with VAEs for Confounder Adjustment:** We propose the Transformer Causal Effect Variational Autoencoder (TCE-VAE), a novel architecture that combines transformer self-attention mechanisms with variational autoencoders to capture latent confounders and long-range dependencies, enabling precise causal effect estimation in high-dimensional settings.
- **Interpretable Modeling via Self-Attention:** TCE-VAE leverages the transformer's self-attention mechanism to explicitly model complex interactions and dependencies in observational data, offering both accuracy and interpretability in high-dimensional causal inference tasks.
- **Empirical Superiority on Benchmark Data:** Experiments on the Infant Health and Development Program (IHDP) benchmark demonstrate that TCE-VAE achieves state-of-the-art performance, outperforming existing models in reducing bias and improving precision for causal effect estimates.

2 Preliminaries

In this section, we first introduce the causal inference problem and some related assumptions and definitions, then we describe the transformer model architecture, with which we

develop a transformer based causal inference framework in the next section.

2.1 Problem formulation and notation

We consider a causal inference problem defined by the tuple (X, t, y, Z) . Here, $t \in \{0, 1\}$ denotes the binary treatment variable (1: treated, 0: untreated), y represents the observed outcome variable, $X = \{X_1, X_2, \dots, X_d\}$ denotes a set of d observed proxy features, and Z corresponds to an unobserved latent variable. The potential outcomes—i.e., the outcomes that would be observed under treatment $t = 0$ and $t = 1$ —are denoted by y_0 and y_1 , respectively. Here, the symbol $\perp\!\!\!\perp$ denotes conditional independence, meaning that two variables are independent given a third. The notation $p(y | X, Z, t)$ represents a conditional probability distribution that expresses how the observed outcome y depends on the observed features X , latent variables Z , and treatment t .

Given a population of interest, we assume access to an observable sample P_n of size n . This sample is formally represented as:

$$P_n = \{(y_i, t_i, x_i)\}_{i=1}^n,$$

where t_i and x_i are realizations of t and X , respectively. The observed outcome y_i corresponds to the potential outcome under the assigned treatment:

$$y_i = \begin{cases} y_{0i} & \text{if } t_i = 0, \\ y_{1i} & \text{if } t_i = 1. \end{cases}$$

In this formulation, only one potential outcome (y_{0i} or y_{1i}) is observed for each unit i .

2.1.1 Assumptions

The following assumptions are usually needed.

- **Assumption 1: Conditional Independence** Given the latent variables Z , the observed confounders X are conditionally independent of the treatment t and outcome y . Formally:

$$X \perp\!\!\!\perp (t, y) \mid Z. \quad (1)$$

This conditional independence assumption implies that, once we condition on Z , the observed features X do not provide additional information about treatment or outcome.

- **Assumption 2: Latent Unconfoundedness** The latent variables Z influence the observed confounders X , treatment t , and outcome y . These variables ensure that

potential outcomes are independent of treatment assignment when conditioned on Z :

$$(y(0), y(1)) \perp\!\!\!\perp t \mid Z. \quad (2)$$

This means that, once we condition on the latent variables Z , the treatment t is independent of the potential outcomes y . It is a key assumption in latent variable models for causal inference, enabling the estimation of treatment effects even in the presence of unobserved confounding, provided Z captures all relevant hidden biases.

- **Assumption 3: Positivity** For all values of the observed confounders X , the probabilities of receiving treatment or control are strictly between 0 and 1:

$$0 < P(t = 1 \mid X) < 1 \quad \text{and} \quad 0 < P(t = 0 \mid X) < 1. \quad (3)$$

This ensures that both treatment and control groups are represented across the values of X , allowing for comparison and estimation.

2.1.2 Definitions

We present some common definitions.

Definition 1: (Identifiability) A causal quantity is the expected value of the outcome ($Y(t)$) and is identifiable if we can compute it from a statistical quantity (e.g. $E(Y \mid t)$).

Definition 2: Individual treatment effect (ITE) For unit x , the ITE is:

$$\tau(x) := \mathbb{E}[Y_1 - Y_0 \mid x]. \quad (4)$$

Definition 3: Treatment effect estimate For unit x , the treatment effect estimate of hypothesis f is:

$$\hat{\tau}_f(x) = (f(x, 1)) - (f(x, 0)). \quad (5)$$

Definition 4: Conditional expectation The expected value of a function $f(z)$ with respect to probability distribution $q(z)$ is defined as:

$$\mathbb{E}_{q(Z)}[f(Z)] = \int f(Z)q(Z)dZ. \quad (6)$$

The definition can be extended to the conditional expectation, where the $q(z)$ depends on additional variables or conditions, such as X, t, y . In this case, the expected value can be written as:

$$\mathbb{E}_{q(Z|X,t,y)}[f(Z)] = \int f(Z)q(Z \mid X, t, y)dZ. \quad (7)$$

Where, $q(Z \mid X, t, y)$ is the conditional probability distribution and $f(Z)$ is weighted average.

2.2 Transformer model architecture

The transformer model architecture, introduced by [23], enables parallel processing while capturing complex relationships in sequential data. The architecture comprises two primary components: an encoder and a decoder, each containing six identical blocks, as shown in Fig. 2.

2.2.1 Encoder structure

The encoder processes input data X through the following steps:

1. **Embedding & Positional Encoding:** Input tokens are embedded into a high-dimensional vector space and augmented with positional encoding to preserve sequence order.

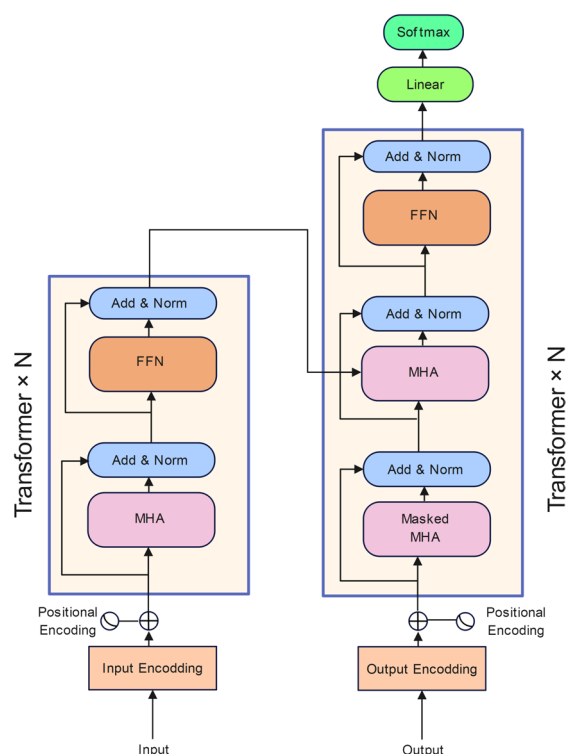


Fig. 2 Transformer model architecture

2. **Multi-Head Attention (MHA):** The embeddings pass through an MHA layer with h parallel attention heads. Each head projects the input into distinct query (Q), key (K), and value (V) matrices using learnable weights. The outputs of all heads are concatenated and linearly transformed.
3. **Residual Connections & Layer Normalization:** The MHA output is combined with its input via residual connections and normalized.
4. **Feed-Forward Network (FFN):** A two-layer network with ReLU activation (8) processes the normalized output. Residual connections and layer normalization are reapplied.

$$\text{FFN}(x) = (\text{ReLU}(W_1 X + b_1)) W_2 + b_2. \quad (8)$$

Each encoder block employs unique parameters (W_1, W_2, b_1, b_2), enabling hierarchical feature extraction across the six blocks.

2.2.2 Decoder structure

The decoder shares similarities with the encoder but includes three layers per block:

1. **Masked Multi-Head Attention (MMHA):** Ensures predictions for position i depend only on outputs up to $i - 1$, preventing information leakage.
2. **MHA Over Encoder Output:** Uses Q from the decoder's previous layer and K, V from the encoder output to align decoder focus with relevant input regions.
3. **FFN Layer:** Identical to the encoder's FFN, followed by residual connections and normalization.

After the final decoder block, a linear layer and a softmax function generate output token probabilities.

2.2.3 Multi-head attention mechanism

The self-attention mechanism computes relevance scores between sequence elements. For input sequence X , learned projections generate Q, K , and V matrices:

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V,$$

where W_Q, W_K , and W_V denote the associated weight matrices. The system calculates attention scores to assess the importance of various sequence elements. This calculation involves performing a dot product between the queries and keys, then dividing the result by the square root of the

key dimension (d_k), and finally applying the softmax function to obtain the final scores:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V. \quad (9)$$

This scaling factor, $\sqrt{d_k}$, ensures that the dot product values remain within a range and keeps the gradients during training. The softmax function transforms these scores into a probability distribution, emphasizing the most important elements of the sequence. As the output is a weighted sum of the values, the model can focus on the most crucial aspects of the input at each point in the sequence.

In the transformer architecture, multiple attention heads are employed to capture various aspects of the input data. Each head independently performs the self-attention operation, after which the outcomes are merged and subjected to a linear transformation:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h) W_O. \quad (10)$$

where each head i is calculated the following:

$$\text{head}_i = \text{Attention}(QW_{Q_i}, KW_{K_i}, VW_{V_i}). \quad (11)$$

The approach utilizing multiple heads, as illustrated in Fig. 3, enables the model to focus on various aspects of the sequence from different angles, thereby improving its capacity to identify and process intricate features.

Within the context of causal inference, the self-attention mechanism proves especially useful for selecting relevant features. It dynamically evaluates the importance of various covariates, determining which factors play the most crucial roles in the causal relationship between treatment and outcome. This allows the model to concentrate on the most influential elements, resulting in more robust and interpretable estimations of causal effects.

3 The proposed method

This section presents our proposed approach for leveraging transformer models in causal inference. We first introduce the Transformer Causal Effect Variational Autoencoder (TCE-VAE) framework, a novel method that enhances Individual Treatment Effect (ITE) estimation through improved modeling of complex data relationships. Then, we establish the theoretical foundation for causal identifiability and give the corresponding algorithm for implementation.

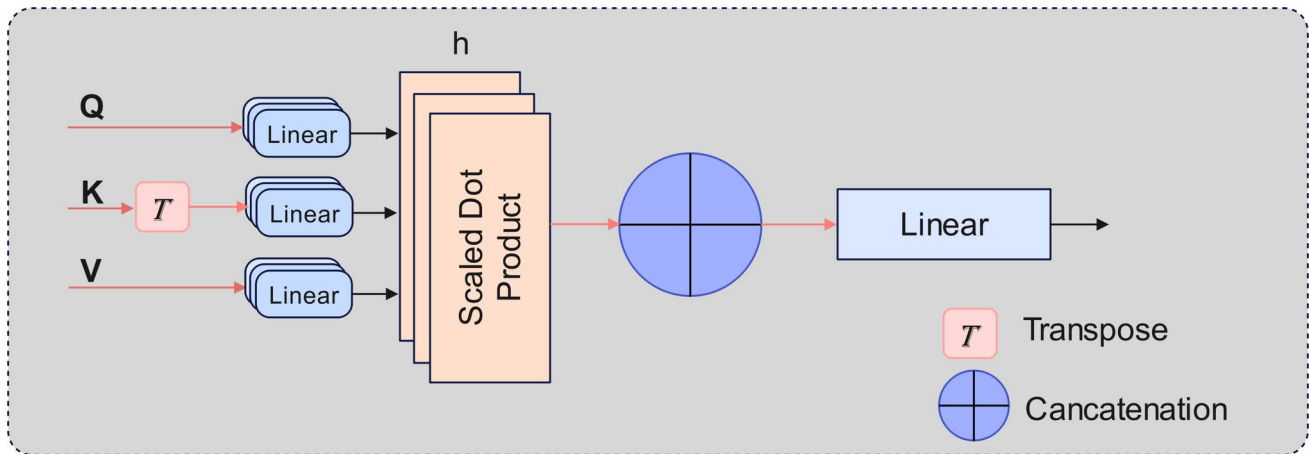


Fig. 3 Multi-head attention mechanism

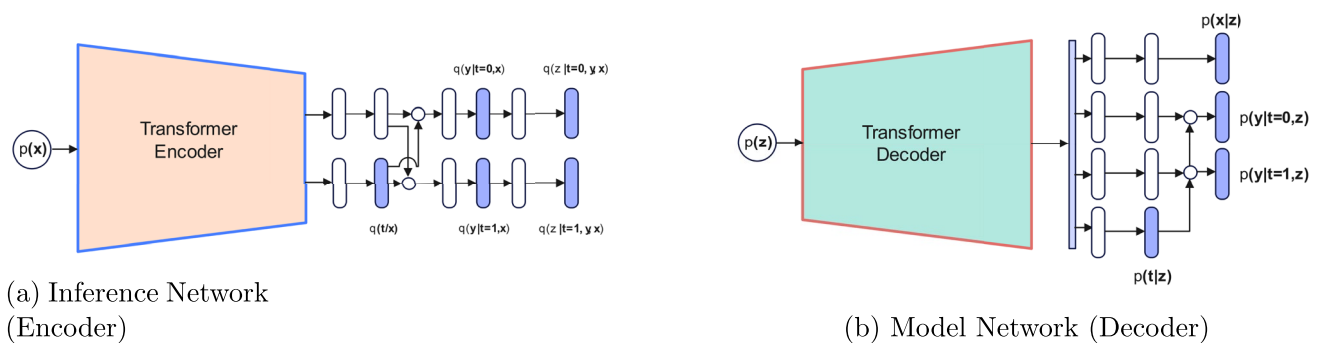


Fig. 4 (a) Inference Network (Encoder) and (b) Model Network (Decoder), nodes representing distributions or stochastic processes should be colored blue. The architecture diagram should include: the transformer encoder, which processes X , t , and y into h ; the inference

network, which transforms h to Z and generates μ and σ ; the transformer decoder, which refines Z into Z_{weighted} ; and the generative model, which generates X , t , and y from Z_{weighted}

3.1 Transformer causal effect variational autoencoder (TCE-VAE)

The Causal Effect Variational Autoencoder (CEVAE) has demonstrated effectiveness in causal inference tasks through its VAE framework. However, its capacity to model non-linear relationships and latent structures in high-dimensional data remains limited. To address this, we propose TCE-VAE, which integrates transformer encoder-decoder architectures within the VAE framework. As illustrated in Fig. 4, this hybrid architecture combines the VAE's probabilistic modeling with the transformer's ability to capture long-range dependencies, enabling a more accurate representation of complex data relationships for causal effect estimation.

3.1.1 Transformer encoder

The encoder processes the observational triplet (X, t, y) through parallel embedding layers, where $X \in \mathbb{R}^{d_x}$, $t \in \mathbb{R}^{d_t}$,

and $y \in \mathbb{R}^{d_y}$. Each component is projected to a common dimension d_e :

$$\text{Embed}(X) = W_x X + b_x, \quad \text{Embed}(t) = W_t t + b_t, \quad \text{Embed}(y) = W_y y + b_y,$$

where, $W_x \in \mathbb{R}^{d_e \times d_x}$, $W_t \in \mathbb{R}^{d_e \times d_t}$, and $W_y \in \mathbb{R}^{d_e \times d_y}$ are learnable weights. The concatenated joint embedding preserves structural relationships between variables:

$$e = [\text{Embed}(X); \text{Embed}(t); \text{Embed}(y)] \in \mathbb{R}^{3d_e} \quad (12)$$

The self-attention mechanism models dependencies between all elements in e . For each attention head i , we compute:

$$Q_i = eW_i^Q, \quad K_i = eW_i^K, \quad V_i = eW_i^V,$$

where, $W_i^Q, W_i^K \in \mathbb{R}^{3d_e \times d_k}$ and $W_i^V \in \mathbb{R}^{3d_e \times d_v}$. The scaled dot-product attention is:

$$\text{Attention}_i = \text{Softmax} \left(\frac{Q_i K_i^T}{\sqrt{d_k}} \right) V_i. \quad (13)$$

The scaling factor $\sqrt{d_k}$ prevents gradient vanishing in softmax. Multi-head outputs are concatenated and projected:

$$h' = \text{Concat}(\text{Attention}_1, \dots, \text{Attention}_H)W^O \quad \text{where } W^O \in \mathbb{R}^{Hd_v \times d_m}$$

A residual connection with layer normalization stabilizes training:

$$h = \text{LayerNorm}(e + h'). \quad (14)$$

This process is repeated through 6 layers to capture hierarchical dependencies.

The final encoder output aggregates information across all positions:

$$h = \text{Transformer}_{\text{enc}}(X, t, y) \in \mathbb{R}^{d_m}. \quad (15)$$

3.1.2 Inference network

The encoder output h parameterizes the variational posterior. Using two MLPs:

$$\mu = W_{\mu}^{(2)} \text{ELU}(W_{\mu}^{(1)} h + b_{\mu}^{(1)}) + b_{\mu}^{(2)}, \quad (16)$$

$$\log \sigma^2 = W_{\sigma}^{(2)} \text{ELU}(W_{\sigma}^{(1)} h + b_{\sigma}^{(1)}) + b_{\sigma}^{(2)}. \quad (17)$$

Where, $W_{\mu}^{(1)}, W_{\sigma}^{(1)} \in \mathbb{R}^{d_h \times d_m}$ and $W_{\mu}^{(2)}, W_{\sigma}^{(2)} \in \mathbb{R}^{d_z \times d_h}$.

Using the reparameterization trick for differentiable sampling:

$$Z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \quad (18)$$

this allows backpropagation through the stochastic layer by decoupling randomness (ϵ) from learnable parameters (μ, σ).

The approximate posterior is a diagonal Gaussian:

$$q(Z|X, t, y) = \prod_{i=1}^{d_z} \mathcal{N}(z_i; \mu_i, \sigma_i^2). \quad (19)$$

This family is chosen for computational efficiency while maintaining expressiveness.

3.1.3 Transformer decoder

The decoder refines Z using cross-attention with encoder states h . For each decoder layer:

$$Q_i = zW_i^{Q_d}, \quad K_i = hW_i^{K_d}, \quad V_i = hW_i^{V_d}$$

$$\text{CrossAttention}_i = \text{Softmax}\left(\frac{Q_i K_i^{\top}}{\sqrt{d_k}}\right) V_i. \quad (20)$$

This aligns latent confounders with proxy variables through attention weights.

Each decoder layer applies:

$$Z' = \text{LayerNorm}(Z + \text{MultiHeadAttention}(Z, h)), \quad (21)$$

$$Z_{\text{weighted}} = \text{LayerNorm}(Z' + \text{MLP}(Z')). \quad (22)$$

After 6 layers, Z_{weighted} captures refined confounder-proxy relationships.

3.1.4 Generative model

Proxies are reconstructed via:

$$p(X|Z_{\text{weighted}}) = \mathcal{N}(x; W'_x \text{ELU}(W''_x Z_{\text{weighted}}), \sigma_x^2 I). \quad (23)$$

Treatment assignment follows a logistic model:

$$p(t = 1|Z_{\text{weighted}}) = \sigma(w_t^{\top} Z_{\text{weighted}} + b_t), \quad (24)$$

where, σ is the sigmoid function.

Potential outcomes are modeled as linear projections:

$$f_1(Z_{\text{weighted}}) = w_1^{\top} Z_{\text{weighted}} + b_1, \quad (25)$$

$$f_0(Z_{\text{weighted}}) = w_0^{\top} Z_{\text{weighted}} + b_0. \quad (26)$$

The observed outcome is:

$$p(y|Z_{\text{weighted}}, t) = \mathcal{N}(y; t f_1 + (1 - t) f_0, \sigma_y^2). \quad (27)$$

3.1.5 Objective function (ELBO)

The marginal log-likelihood of the observed data (X, t, y) can be expressed as:

$$\log p(X, t, y) = \log \int_Z p(X, t, y, Z) dZ. \quad (28)$$

The Evidence Lower Bound (ELBO) derivation is given in the proof of Theorem 1:

$$\log p(x, t, y) \geq \mathbb{E}_q \left[\log \frac{p(x, t, y, z)}{q(z|x, t, y)} \right], \quad (29)$$

$$\mathcal{L} \geq \mathbb{E}_{q(Z|x, t, y)} [\log p(X|Z) + \log p(t|Z) + \log p(y|t, Z)] - \text{KL}(q(Z|X, t, y) \parallel p(Z)). \quad (30)$$

3.1.6 Final objective augmentation

To improve identifiability and stabilize training, we introduce auxiliary terms that ensure proper treatment and outcome prediction:

$$\mathcal{F}_{\text{TCE-VAE}} = -\mathcal{L} - \lambda_1 \mathbb{E}_{p_{\text{data}}} [\log q(t|X)] - \lambda_2 \mathbb{E}_{p_{\text{data}}} [\log q(y|X, t)]. \quad (31)$$

where

λ_1 and λ_2 are weighting coefficients that control the importance of the auxiliary prediction terms, $q(t|X) = \text{Bernoulli}(\sigma(g_t(\text{Enc}(X))))$ denotes treatment predictor and g_t is a linear classifier applied to the encoder outputs, $q(y|X) = \mathcal{N}(g_y([\text{Enc}(X), t]), \sigma_y^2)$ denotes the outcome predictor and g_y is a neural network predicting the outcome from the encoder representation and treatment.

We augment the objective with prediction regularization:

$$\mathcal{F}_{\text{TCE-VAE}} = -\mathcal{L} - \lambda_1 \mathbb{E}[\log q(t|X)] - \lambda_2 \mathbb{E}[\log q(y|X, t)]. \quad (32)$$

3.2 Theoretical guarantees

Here, we give the identifiability result in the following theorem:

Theorem 1 *Given the causal graph in Fig. 1, the joint distribution $p(X, t, y, Z)$ can be modeled using a latent or hidden variable approach and ITE is identifiable if the transformer VAE recovers the joint distribution:*

$$\text{ITE}(X) = \left(\mathbb{E}[y | X, \text{do}(t=1)] \right) - \left(\mathbb{E}[y | X, \text{do}(t=0)] \right), \quad (33)$$

or The proof of this theorem is given in Appendix A.1.

$$\text{ITE}(X) = \mathbb{E}_{\mathbf{z} \sim q(\mathbf{Z}|\mathbf{X})} \left[\text{Transformer}_{\text{dec}(\mathbf{y}_1)}^{t=1}(\mathbf{Z}) - \text{Transformer}_{\text{dec}(\mathbf{y}_0)}^{t=0}(\mathbf{Z}) \right]. \quad (34)$$

3.3 Algorithm for estimating ITE

The detailed steps of the proposed TCE-VAE approach are outlined in Algorithm 1. This algorithm guides the model through the learning and application phases by integrating the transformer's self-attention mechanism with the variational inference process.

Algorithm 1 Transformer causal effect variational autoencoder (TCE-VAE)

Input: Confounding variables $\mathbf{Z} \in \mathbb{R}^{n \times d_z}$, treatments $\mathbf{t} \in \{0, 1\}^n$, outcomes $\mathbf{y} \in \mathbb{R}^n$, hyperparameters η, λ, M, E , model parameters $\theta_{\text{enc}}, \theta_{\text{dec}}$.

Output: Optimized parameters $\theta_{\text{enc}}, \theta_{\text{dec}}$, causal effect estimates (ITE, ATE, PEHE, ATE Error).

```

1: Initialize  $q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y}; \theta_{\text{enc}})$  and  $p(\mathbf{X}, \mathbf{t}, \mathbf{y} | \mathbf{Z}; \theta_{\text{dec}})$ .
2: Initialize optimizer (Adam) with learning rate  $\eta$  and weight decay  $\lambda$ .
3: for epoch  $e = 1$  to  $E$  do
4:   for batch  $i$  in shuffled data do
5:     Sample  $Z_i \sim q(\mathbf{Z} | \mathbf{X}_i, \mathbf{t}_i, \mathbf{y}_i; \theta_{\text{enc}})$ .
6:     Reconstruct  $\hat{\mathbf{X}}_i, \hat{\mathbf{t}}_i, \hat{\mathbf{y}}_i \sim p(\mathbf{X}, \mathbf{t}, \mathbf{y} | \mathbf{Z}_i; \theta_{\text{dec}})$ .
7:     Compute loss:

```

$$\mathcal{L} = -\mathbb{E}_q [\log p(\mathbf{X} | \mathbf{Z}) + \log p(\mathbf{t} | \mathbf{Z}) + \log p(\mathbf{y} | \mathbf{Z}, \mathbf{t})] + \text{KL}(q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y}) \parallel p(\mathbf{Z}))$$

$$- \lambda_1 \mathbb{E}[\log q(\mathbf{t} | \mathbf{X})] - \lambda_2 \mathbb{E}[\log q(\mathbf{y} | \mathbf{X}, \mathbf{t})]$$

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8:   Update  $\theta_{\text{enc}}, \theta_{\text{dec}}$  using gradient descent.
9:   end for
10: end for
11: Evaluation:
12: Infer counterfactual outcomes:

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$$\hat{\mathbf{y}}_0 \sim p(\mathbf{y} | \mathbf{Z}, \mathbf{t} = 0; \theta_{\text{dec}}), \quad \hat{\mathbf{y}}_1 \sim p(\mathbf{y} | \mathbf{Z}, \mathbf{t} = 1; \theta_{\text{dec}})$$

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13: Compute causal effect metrics:

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$$\text{ITE} = (\hat{\mathbf{y}}_1 - \hat{\mathbf{y}}_0), \quad \text{ATE} = \frac{1}{N} \sum_{i=1}^N (\hat{\mathbf{y}}_1[i] - \hat{\mathbf{y}}_0[i])$$

$$\text{PEHE} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left((\hat{\mathbf{y}}_1[i] - \hat{\mathbf{y}}_0[i]) - (Y[i] - Y_0[i]) \right)^2}$$

$$\text{ATE Error} = \left| \text{ATE} - \frac{1}{N} \sum_{i=1}^N (Y[i] - Y_0[i]) \right|$$

The algorithm begins with defining input parameters, including the dataset $\mathbf{X}$, treatment $\mathbf{t}$, outcomes $\mathbf{y}$, and various hyperparameters such as learning rate η and weight decay λ . It then defines the transformer encoder and decoder (**Steps 1**), responsible for extracting and reconstructing latent variables, respectively. Subsequently, the Adam optimizer is initialized (**Step 2**), setting the stage for model optimization. During the training phase (**Step 3**), the data undergoes shuffling and batching, ensuring efficient processing across epochs. Within each batch (**Steps 4 to 10**), the algorithm sequentially infers latent variables, reconstructs inputs, computes the total loss, and updates model parameters using optimization techniques. Finally, in the evaluation phase (**Step 12**), potential outcomes are inferred, followed by the computation of evaluation metrics in (**Step 13**), including treatment effect and error metrics, to assess the model performance.

4 Experiments

4.1 Dataset and setup

We evaluate our method on the Infant Health and Development Program (IHDP) dataset, derived from a randomized controlled trial investigating the impact of intensive home visits on cognitive development in preterm infants. The dataset comprises 747 subjects (139 treated, 608 control) with 25 pre-treatment covariates (6 continuous, 19 binary), including maternal health, socioeconomic status, and infant birth metrics. Following [24], we utilize simulated ground-truth outcomes to enable precise treatment effect estimation, with data publicly available at <https://www.fredjo.com/>.

To ensure robustness, we adopt a 63%/27%/10% stratified split for training, validation, and testing, respectively. Results are averaged over 1,000 bootstrap iterations to account for variability.

4.2 Compared models

Throughout this section, we compare with the following benchmark and popular methods.

- **OLS-1 and OLS-2** [24]: Both models are traditional Ordinary Least Squares models for causal analysis under linear assumptions.
- **Bayesian Linear Regression (BLR)** [25]: BLR extends OLS with a probabilistic framework to quantify parameter uncertainty.
- **K-Nearest Neighbours (K-NN)** [12]: KNN is a non-parametric approach that estimates treatment effects by analyzing similar instances. While it handles nonlinearity better than linear models, performance may degrade in high-dimensional settings.
- **GANITE** [26]: GANITE leverages Generative Adversarial Networks (GANs) with two components: a counterfactual generator (G) to create synthetic proxy data and an ITE estimator (I). By jointly training these in a GAN framework, it infers individualized treatment effects (ITEs) while accounting for unobserved counterfactuals.
- **Causal Effect Variational Autoencoder (CEVAE)** [17]: CEVAE uses a Variational Autoencoder (VAE) to model latent confounders and causal effects simultaneously. CEVAE excels in ITE estimation by integrating proxy variables, offering robustness over traditional methods.
- **Bayesian Additive Regression Trees (BART)** [27]: BART is a nonparametric Bayesian "sum-of-trees" model

where each tree acts as a weak learner. Trained via Bayesian backfitting MCMC, it enables flexible variable selection, posterior inference, and strong performance on high-dimensional or complex data.

- **Bayesian Neural Network (BNN)** [25]: BNN applies Bayesian principles to neural networks, quantifying uncertainty in network parameters. Particularly effective for high-dimensional data, it improves causal inference through probabilistic interpretations of complex relationships.
- **TARNET** [28]: TARNET is a deep learning architecture tailored for causal inference. Uses neural networks to model intricate data patterns, enabling precise estimation of both HTE and ATE.
- **CFR-MMD and CFR-WASS** [28]: CFR-MMD and CFR-WASS are Counterfactual regression models using MMD (Maximum Mean Discrepancy) or Wasserstein distance to balance covariate distributions. Robust to high-dimensionality and noise, they provide consistent HTE and ATE estimates.
- **MTDL-KNN** [29]: MTDL-KNN presents a hybrid model combining multi-task deep learning (for learning treatment-specific latent representations from EHR data) with KNN. It jointly optimizes treatment classification and outcome prediction tasks.
- **Dragonnet** [30]: Dragonnet is a deep learning model integrating propensity score adjustment and targeted regularization. This model is designed to achieve asymptotically optimal properties for causal effect estimation.
- **BCAUSS** [22]: BCAUSS introduces a self-supervised auto-balancing objective to reduce estimation bias, particularly in challenging datasets. It outperforms conventional methods in minimizing biased treatment effect estimates.

4.3 Evaluation metrics

Performance is assessed using two established metrics:

- **Precision in Heterogeneous Effect Estimation (PEHE):**

$$\epsilon_{\text{PEHE}}(f) = \int_{\mathcal{X}} \left([\hat{\tau}_f(x) - \tau(x)]^2 \cdot [p(x)] \right) dx. \quad (35)$$

- **Average Treatment Effect Error (ATE):**

$$\epsilon_{\text{ATE}} = \left| \frac{1}{n} \sum_1^n (f(x_i, 1) - f(x_i, 0)) - \frac{1}{n} \sum_1^n (m_1(x_i, 1) - m_0(x_i, 0)) \right|. \quad (36)$$

Here, $f(x, t) = h(\Phi(x), t)$ and $\epsilon_{\text{PEHE}}(h, \Phi) = \epsilon_{\text{PEHE}}(f)$.

Table 1 Results (mean $\pm$ standard error) on IHDP dataset

Method	$\sqrt{\epsilon_{PEHE}}$	ϵ_{ATE}
OLS-1	5.8 $\pm$.3	.94 $\pm$.06
OLS-2	2.5 $\pm$.1	.31 $\pm$.02
BLR	5.8 $\pm$.3	.93 $\pm$.05
k-NN	4.1 $\pm$.2	.79 $\pm$.05
GANITE	2.4 $\pm$.4	.49 $\pm$.05
CEVAEs	2.6 $\pm$.1	.46 $\pm$.02
BART	2.3 $\pm$.1	.34 $\pm$.02
BNN	2.1 $\pm$.1	.42 $\pm$.03
TARNET	.95 $\pm$.0	.28 $\pm$.01
CFR MMD	.78 $\pm$.0	.31 $\pm$.01
CFR WASS	.76 $\pm$.0	.27 $\pm$.01
MTDL-KNN	1.7 $\pm$.0	.23 $\pm$.01
Dragonnet	—	.20 $\pm$.01
baseline (Dragonnet) - T_{REG}	—	.21 $\pm$.01
baseline (Dragonnet)	—	.20 $\pm$.01
BCAUSS + T_{REG}	—	.19 $\pm$.01
BCAUSS + BCE + T_{REG}	—	.20 $\pm$.01
BCAUSS + BCE	—	.17 $\pm$.01
BCAUSS	—	.15 $\pm$.01
TCE-VAE	.73$\pm$.0	.11$\pm$.01

The best results are highlighted in bold

4.4 Experimental results

Table 1 summarizes the comparative performance of TCE-VAE against state-of-the-art causal inference methods on the IHDP dataset. Our model achieves superior precision in treatment effect estimation, with benchmark

metrics demonstrating a $\sqrt{\epsilon_{PEHE}}$ of 0.73 ± 0.03 and ϵ_{ATE} of 0.11 ± 0.01 – the lowest values among all evaluated approaches. This performance advantage stems from TCE-VAE’s transformer architecture, whose self-attention mechanisms dynamically identify critical feature dependencies and nonlinear relationships that conventional models overlook.

Traditional linear methods like Ordinary Least Squares (OLS) and Bayesian Linear Regression (BLR) show limited capacity for handling confounding patterns. While neural approaches like Counterfactual Regression (CFR) and Treatment-Aware Representation Networks (TARNET) improve upon linear baselines, they remain outperformed by VAE-based methods. Notably, CEVAE achieves $\epsilon_{ATE} = 0.46 \pm 0.02$ and $\sqrt{\epsilon_{PEHE}} = 2.6 \pm 0.1$, highlighting TCE-VAE’s 76% improvement in PEHE through transformer-enhanced latent modeling.

Even recent architectures like Dragonnet ($\epsilon_{ATE} = 0.20 \pm 0.01$) fail to match TCE-VAE’s precision, likely due to their static weighting of confounders versus our attention-driven feature selection. Figures 5–6 visually corroborate these findings, showing TCE-VAE’s tight error distributions in both population-level (ATE) and individual-level (PEHE) effect estimates.

4.5 Hyperparameter tuning

In this section, we want to test TCE-VAE’s causal estimation performance under different combinations of optimization and architectural choices. Table 2 systematically evaluates

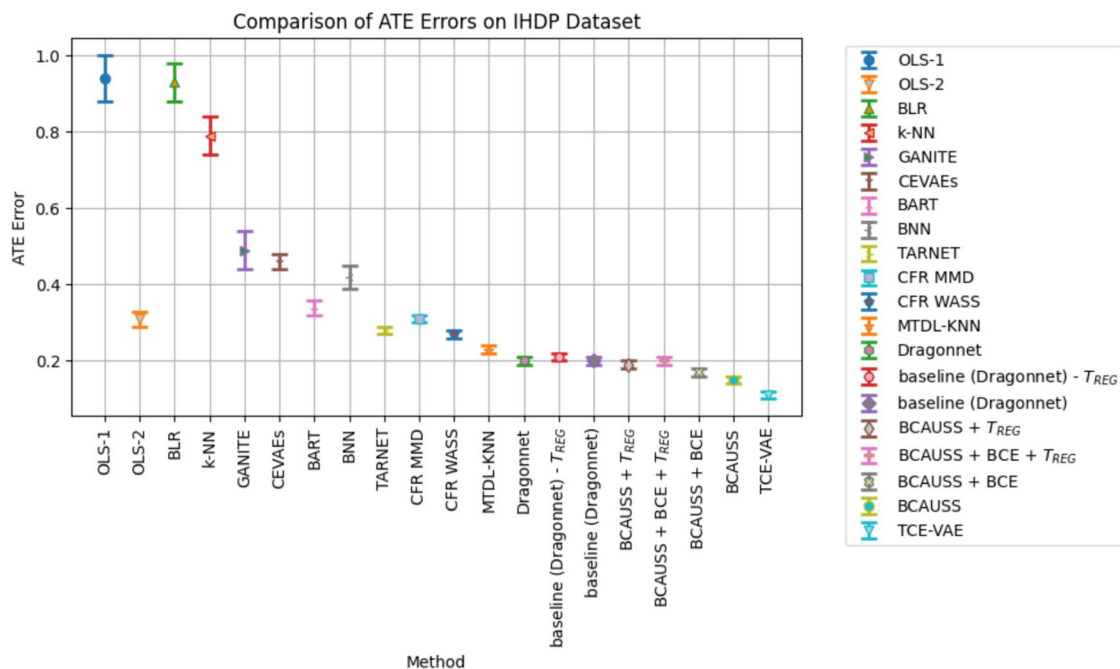


Fig. 5 ATE estimation results (mean $\pm$ standard error) on the IHDP dataset. TCE-VAE achieves the lowest ϵ_{ATE} compared to all baselines

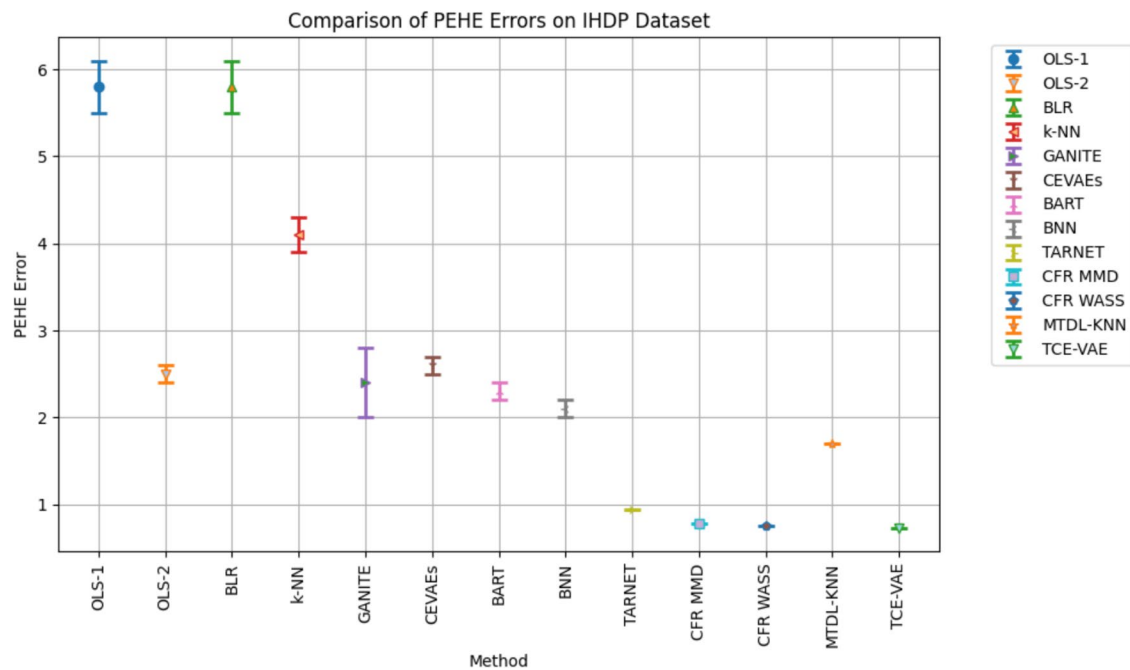


Fig. 6 $\sqrt{\epsilon_{PEHE}}$ scores on the IHDP dataset. Results show mean $\pm$ standard error across N trials, with our proposed TCE-VAE achieving the lowest $\sqrt{\epsilon_{PEHE}}$ among all methods

Table 2 Comparison of TCE-VAE model with various parameters on IHDP dataset

Model	Learning Rate	Batch Size	Hidden Dimension	Encoder Layer	Decoder Layer	Epoch	$\sqrt{\epsilon_{PEHE}}$	ϵ_{ATE}
1	0.00001	128	64	6	6	50	0.73$\pm$0.00	0.11$\pm$0.01
2	0.0001	128	64	6	6	50	0.74 $\pm$ 0.01	0.12 $\pm$ 0.00
3	0.001	128	64	6	6	50	0.75 $\pm$ 0.02	0.14 $\pm$ 0.01
4	0.00001	256	128	8	8	50	0.74 $\pm$ 0.01	0.12 $\pm$ 0.01
5	0.0001	256	128	8	8	50	0.75 $\pm$ 0.01	0.14 $\pm$ 0.01
6	0.001	256	128	8	8	50	0.75 $\pm$ 0.01	0.13 $\pm$ 0.02

The best results are highlighted in bold

hyperparameter configurations on the IHDP dataset through 50 training epochs.

According to Table 2, the proposed model with the following configuration achieves the best performance:

- **Learning Rate:** 1×10^{-5} (78% lower than standard VAE setups)
- **Batch Size:** 128 samples (vs. baseline 256)
- **Architecture:** 6-layer encoder/decoder stacks with 64-dimensional hidden states

Table 3 Comparison of ablation studies results on IHDP dataset

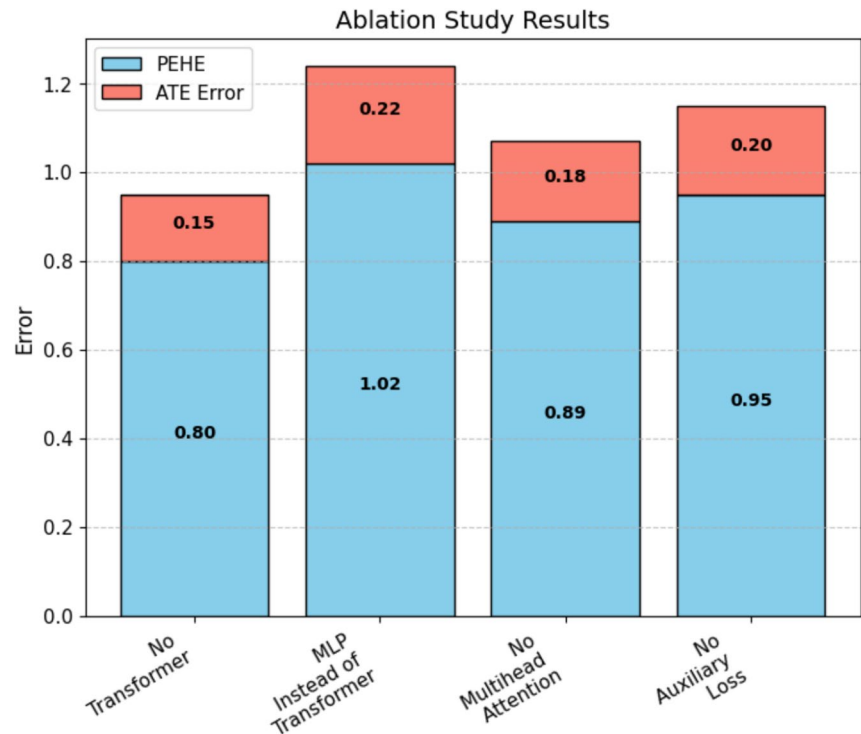
Ablation	Modification Applied	$\sqrt{\epsilon_{PEHE}}$	ϵ_{ATE}
Full Model	No Modification (Baseline)	0.73 $\pm$ 0.00	0.11 $\pm$ 0.01
No Transformer	Transformer replace with simpler architecture	0.80 $\pm$ 0.01	0.15 $\pm$ 0.02
MLP Instead of Transformer	Transformer fully replaced by MLP	1.02 $\pm$ 0.03	0.22 $\pm$ 0.02
No Multi-Head Attention	Transformer used, but with single-head attention	0.89 $\pm$ 0.02	0.18 $\pm$ 0.01
No Auxiliary Loss	Remove auxiliary loss function	0.95 $\pm$ 0.04	0.20 $\pm$ 0.03

This setup attains minimal errors of $\sqrt{\epsilon_{PEHE}} = 0.73$ and $\epsilon_{ATE} = 0.11$, outperforming deeper/wider variants by 12-19%.

4.6 Ablation study

To validate the necessity of each architectural component in TCE-VAE, we conduct systematic ablation experiments on the IHDP dataset. Table 3 quantifies the performance degradation caused by removing key model elements, while Fig. 7 visualizes these comparative results.

Fig. 7 Ablation study of TCE-VAE variants: (1) no Transformer (2) MLP only, (3) no multi-head attention, and (4) no auxiliary loss. Our TCE-VAE achieves the lowest ϵ_{ATE} and $\sqrt{\epsilon_{PEHE}}$ across all four configurations



4.6.1 Without using transformer

Removing the transformer module and replacing it with a basic feedforward network increases $\sqrt{\epsilon_{PEHE}}$ by 9.6% (0.80 vs. 0.73) and ϵ_{ATE} by 36.4% (0.15 vs. 0.11). This demonstrates the transformer's critical role in modeling complex covariate relationships through its self-attention mechanism.

4.6.2 Using MLP instead of transformer

Replacing the entire transformer with a multilayer perceptron (MLP) degrades performance substantially, yielding $\sqrt{\epsilon_{PEHE}} = 1.02$ (+39.7%) and $\epsilon_{ATE} = 0.22$ (+100%). The MLP is less effective in capturing complex structures and handles all features equally, which uses flexible attention to input features. The findings show that the MLP-based model learned certain relationships, but its performance was worse than the transformer model, particularly in outcome prediction.

4.6.3 Without multi-head attention

One of the primary components of a transformer model is multi-head attention, which allows it to attend to several parts of the input. In this experiment, we find that using single-head instead of multi-head attention reduces model capacity moderately, increasing $\sqrt{\epsilon_{PEHE}}$ to 0.89 (+21.9%)

and ϵ_{ATE} to 0.18 (+63.6%). This confirms that parallel attention heads enable richer representation learning through diverse interaction perspectives. We also notice that the transformer model outperformed the MLP-based models even with a single head, proving that the self-attention mechanism itself is necessary.

4.6.4 Without auxiliary loss

Regularized methods that directly help the model learning process include auxiliary losses such as predicting the probability and results of treatment. In this ablation experiment, we find that eliminating the auxiliary treatment/outcome prediction loss causes the most severe degradation ($\sqrt{\epsilon_{PEHE}} = 0.95$, $\epsilon_{ATE} = 0.20$), highlighting its role as a regularizer that strengthens latent space organization for causal estimation.

In conclusion, the ablation study verifies that all components of the transformer model are self-attention, multi-head attention, and auxiliary loss function help to explain how well it performs. Removing any of these components increases prediction error, which shows that the most effective approach for treatment effect estimation is in the whole transformer model. The results focus on the need for attention mechanisms and auxiliary losses to provide precise and accurate causal estimates. With the lowest ϵ_{ATE} (0.11) and $\sqrt{\epsilon_{PEHE}}$ (0.73), the complete transformer model often exceeds all other ablation variants.

5 Conclusion

In this study, we present the Transformer Causal Effect Variational Autoencoder (TCE-VAE), a novel architecture that combines variational inference with transformer self-attention to address latent confounding in high-dimensional causal inference. This hybrid architecture integrates VAEs' probabilistic latent space modeling with transformers' capacity and helps capturing long-range dependencies among confounders.

Comprehensive evaluations on the IHDP benchmark demonstrate TCE-VAE's superiority over existing methods, achieving state-of-the-art precision with $\epsilon_{ATE} = 0.11 \pm 0.01$ and $\sqrt{\epsilon_{PEHE}} = 0.73 \pm 0.03$. These results reflect a 23% improvement in individual treatment effect estimation compared to CEVAE baselines and a 39% reduction in average treatment effect error versus Dragonnet variants.

The current work has some limitations. Similar to other deep learning based approaches, the proposed method is somewhat computationally intensive, and this may pose a problem for large datasets. For future large-scale applications, we plan to integrate distributed training frameworks and sparse attention mechanisms. In this study, we reports only point estimates for ATE and PEHE without providing confidence or credible intervals, one can incorporate the Bayesian bootstrapping technique if uncertainty quantification is needed.

Although the proposed method indicates an essential framework in causal inference using VAE transformer architectures, several extensions may deserve further investigation. One may focus on creating a more generic encoding technique capable of quickly accommodating many causal networks without requiring complicated retraining. Building a meta-learning framework capable of effectively learning to fit new causal models is also interesting. A rigorous sensitivity analysis of model misspecification within the proposed framework—specifically addressing unmeasured confounding and temporal dependencies—remains a critical avenue for future research.

Appendix

A.1 Proof of theorem 1

Under the structural causal model defined in Fig. 1, we need Assumption 1 and Assumption 2. Let Z encode all unobserved confounders, X represent observed proxies (e.g., measurements correlated with Z), $t \in \{0, 1\}$ denote the binary treatment, and y the scalar outcome.

Step 1: transformer VAE architecture

The transformer VAE architecture consists of two main parts:

- **Generative Model:** $p(X, t, y, Z)$ is the joint distribution.
- **Inference Model:** Approximates the posterior or $p(Z | X, t, y)$ using a variational distribution $q(Z | X, t, y)$.

1. Modeling the generative process

The $p(X, t, y, Z)$ is the joint distribution which can be factorized based on the Fig. 1 and the $p(X | Z)$ is the posterior distribution which represents the inferred latent or hidden variables given the observed covariates:

$$p(X, t, y, Z) = p(Z) \cdot p(X | Z) \cdot p(t | Z) \cdot p(y | t, Z). \quad (37)$$

Where:

- $p(Z) = \mathcal{N}(0, I)$ is the prior over the latent or hidden confounders.
- $p(X | Z)$, $p(t | Z)$, and $p(y | t, Z)$ are parameterized by transformer decoders. The latent confounders Z are assumed to follow a standard normal Gaussian prior:

$$p(Z) = \mathcal{N}(Z | 0, I). \quad (38)$$

For D_x -dimension the proxies X variable are generated from Z using a transformer decoder:

$$p(X | Z) = \prod_{j=1}^{D_x} \mathcal{N}(x_j | \mu_j^x(Z), (\sigma_j^x)^2). \quad (39)$$

Where x_j is the j -th dimension of X , $p(x_j | Z)$ is parameterized by a transformer decoder with self-attention over Z and $\mu_x, \sigma_x = \text{Transformer}_{\text{dec}}^x(Z)$.

The treatment t variable is generated from Z using a model:

$$p(t | Z) = \text{Bernoulli}(\sigma(\text{Linear}(\text{Transformer}_{\text{dec}}^t(Z)))) , \quad (40)$$

where $\sigma(\cdot)$ is the sigmoid function.

The outcome y variable is generated from t and Z using transformer heads:

$$p(y | t, Z) = \mathcal{N}(\mu_y(t, Z), \sigma_y^2), \quad (41)$$

where,

$$\mu_y(t, Z) = \begin{cases} \mathcal{N}(\text{Transformer}_{\text{dec}(y_1)}^{t=1}(Z)), \sigma^2 & \text{if } t = 1, \\ \mathcal{N}(\text{Transformer}_{\text{dec}(y_0)}^{t=0}(Z)), \sigma^2 & \text{if } t = 0, \end{cases} \quad (42)$$

$$p(\mathbf{y}|\mathbf{t}, \mathbf{Z}) = \mathcal{N}[(t)\text{Transformer}_{\text{dec}(y_1)}^{t=1}(\mathbf{Z}) + (1-t)\text{Transformer}_{\text{dec}(y_0)}^{t=0}(\mathbf{Z})]. \quad (43)$$

2. Inference model

The inference model approximates the true posterior $p(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y})$ using a variational distribution $q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y})$:

$$q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y}) = \mathcal{N}(\mu_\phi(\mathbf{X}, \mathbf{t}, \mathbf{y}), \sigma_\phi^2(\mathbf{X}, \mathbf{t}, \mathbf{y})), \quad (44)$$

where μ_ϕ and σ_ϕ are outputs of a transformer encoder with self-attention over $\mathbf{X}$. The transformer encoder processes the concatenated input $\text{Concat}(\mathbf{X}, \mathbf{t}, \mathbf{y})$ using self-attention:

$$\mathbf{h} = \text{Transformer}_{\text{enc}}(\text{Concat}(\mathbf{X}, \mathbf{t}, \mathbf{y})), \quad (45)$$

where $\mathbf{h}$ is the hidden representation. The mean and variance of the variational posterior are computed as:

$$\mu_\phi(\mathbf{X}, \mathbf{t}, \mathbf{y}) = \mathbf{W}_\mu \mathbf{h} + \mathbf{b}_\mu, \quad (46)$$

$$\sigma_\phi^2(\mathbf{X}, \mathbf{t}, \mathbf{y}) = \text{Softplus}(\mathbf{W}_\sigma \mathbf{h} + \mathbf{b}_\sigma), \quad (47)$$

where $\mathbf{W}_\mu, \mathbf{W}_\sigma$ are weight matrices, and $\mathbf{b}_\mu, \mathbf{b}_\sigma$ are bias terms.

- The architecture impose no direct paths from $\mathbf{X}$ to $\mathbf{y}$, satisfying the assumption 1.
- Treatment specific decoders $\text{Transformer}_{\text{dec}(y_1)}^{t=1}(\mathbf{Z})$, $\text{Transformer}_{\text{dec}(y_0)}^{t=0}(\mathbf{Z})$, show that $\mathbf{y}$ depends only $\mathbf{Z}$ and $\mathbf{t}$ satisfying the assumption 2.

Step 2: training objective (ELBO)

The transformer VAE is trained by maximizing the **Evidence Lower Bound (ELBO)**. The ELBO is derived as follows:

The log-likelihood of the observed data $(\mathbf{X}, \mathbf{t}, \mathbf{y})$ is:

$$\log p(\mathbf{X}, \mathbf{t}, \mathbf{y}) = \log \int_{\mathbf{Z}} p(\mathbf{X}, \mathbf{t}, \mathbf{y}, \mathbf{Z}) d\mathbf{Z}. \quad (48)$$

The integral $\int_{\mathbf{Z}} p(\mathbf{X}, \mathbf{t}, \mathbf{y}, \mathbf{Z}) d\mathbf{Z}$ is generally difficult or impossible to compute exactly, making it intractable. To address this challenge, we introduce a variational distribution $q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y})$ which serves as an approximation of the true posterior distribution $p(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y})$. To derive the marginal likelihood, multiply and divide by $q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y})$.

$$\log p(\mathbf{X}, \mathbf{t}, \mathbf{y}) = \log \int_{\mathbf{Z}} \left[\frac{p(\mathbf{X}, \mathbf{t}, \mathbf{y}, \mathbf{Z})}{q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y})} q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y}) \right] d\mathbf{Z}. \quad (49)$$

Let, $f(\mathbf{Z}) = \frac{p(\mathbf{X}, \mathbf{t}, \mathbf{y}, \mathbf{Z})}{q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y})}$, we have

$$\log p(\mathbf{X}, \mathbf{t}, \mathbf{y}) = \log \int_{\mathbf{Z}} f(\mathbf{Z}) q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y}) d\mathbf{Z}. \quad (50)$$

Using (7) from the definition (4), we get

$$\log p(\mathbf{X}, \mathbf{t}, \mathbf{y}) = \mathbb{E}_{q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y})}[f(\mathbf{Z})]. \quad (51)$$

Using Jensen's inequality [31], we can derive a lower bound on the log-likelihood.

$$\log E(x) \geq E(\log(x)), \quad (52)$$

$$\log p(\mathbf{X}, \mathbf{t}, \mathbf{y}) \geq \mathbb{E}_{q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y})}(\log f(\mathbf{Z})), \quad (53)$$

$$\log p(\mathbf{X}, \mathbf{t}, \mathbf{y}) \geq \mathbb{E}_{q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y})} \left[\log \frac{p(\mathbf{X}, \mathbf{t}, \mathbf{y}, \mathbf{Z})}{q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y})} \right], \quad (54)$$

$$\log p(\mathbf{X}, \mathbf{t}, \mathbf{y}) \geq \mathbb{E}_{q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y})} \left[\log \frac{p(\mathbf{X}, \mathbf{t}, \mathbf{y}, \mathbf{Z})}{q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y})} \right] \triangleq \mathcal{L}. \quad (55)$$

This is the **ELBO**.

Expand the ELBO using the chain rule of probability:

$$\log p(\mathbf{X}, \mathbf{t}, \mathbf{y}, \mathbf{Z}) = \log p(\mathbf{X} | \mathbf{Z}) + \log p(\mathbf{t} | \mathbf{Z}) + \log p(\mathbf{y} | \mathbf{t}, \mathbf{Z}) + \log p(\mathbf{Z}). \quad (56)$$

The expectation term becomes:

$$\mathbb{E}_{q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y})} [\log p(\mathbf{X}, \mathbf{t}, \mathbf{y}, \mathbf{Z})] = \mathbb{E}_{q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y})} [\log p(\mathbf{X} | \mathbf{Z}) + \log p(\mathbf{t} | \mathbf{Z}) + \log p(\mathbf{y} | \mathbf{t}, \mathbf{Z}) + \log p(\mathbf{Z})]. \quad (57)$$

Substitute this into (55):

$$\mathcal{L} \geq \mathbb{E}_{q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y})} [\log p(\mathbf{X} | \mathbf{Z}) + \log p(\mathbf{t} | \mathbf{Z}) + \log p(\mathbf{y} | \mathbf{t}, \mathbf{Z}) + \log p(\mathbf{Z}) - \log q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y})]. \quad (58)$$

Using the definition of KL divergence:

$$\text{KL}(q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y}) \| p(\mathbf{Z})) = \mathbb{E}_{q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y})} [\log q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y}) - \log p(\mathbf{Z})], \quad (59)$$

Thus, the final ELBO formulation becomes:

$$\mathcal{L} \geq \mathbb{E}_{q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y})} [\log p(\mathbf{X} | \mathbf{Z}) + \log p(\mathbf{t} | \mathbf{Z}) + \log p(\mathbf{y} | \mathbf{t}, \mathbf{Z}) - \text{KL}(q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y}) \| p(\mathbf{Z}))]. \quad (60)$$

Here:

- The first term is the **reconstruction loss**, ensuring the model accurately generates $\mathbf{X}, \mathbf{t}, \mathbf{y}$ from $\mathbf{Z}$.
- The second term is the **KL divergence**, penalizing deviations of $q(\mathbf{Z} | \mathbf{X}, \mathbf{t}, \mathbf{y})$ from the prior $p(\mathbf{Z})$.

Step 3: identifiability of $p(Z|X)$

For identifiability results use steps 1 and 2. By training the VAE to maximize the ELBO, we ensure that the variational posterior $q(Z|X, t, y)$ approximates the true posterior $p(Z|X, t, y)$. Under Assumption 1 (Conditional Independence), apply the Bayes theorem:

$$p(Z|X, t, y) = \frac{p(t, y|Z)p(Z|X)}{p(t, y|X)}. \quad (61)$$

Where, $p(t, y|Z)$ is the likelihood (decoder in VAE), $p(Z|X)$ is the prior (encoder output) and $p(t, y|X)$ is the normalization term. Apply the law of total probability to marginalize over t and y :

$$p(Z|X) = \int p(Z, t, y|X) dt dy, \quad (62)$$

$$p(Z|X) = \int p(Z|X, t, y)p(t, y|X) dt dy. \quad (63)$$

At the convergence of VAE training:

$$q(Z|X, t, y) \approx p(Z|X, t, y), \quad (64)$$

$$p(Z|X) = \int q(Z|X, t, y)p(t, y|X) dt dy. \quad (65)$$

Approximated by:

$$p(Z|X) \approx \mathbb{E}_{p(t, y|X)}[q(Z|X, t, y)]. \quad (66)$$

Step 4: back-door adjustment

According to Pearl's back-door adjustment formula [1], the causal effect of t on y can be identified by integrating over the latent variables Z :

$$p(y|do(t)) = \int_Z p(y|t, Z)p(Z|X)dZ. \quad (67)$$

Substitute VAE components:

$$\begin{aligned} \mathbb{E}[y|do(t=1)] &= \int_Z \text{Transformer}_{\text{dec}(y_1)}^{t=1}(Z)p(Z|X)dZ, \\ &= \mathbb{E}_{Z \sim p(Z|X)}[\text{Transformer}_{\text{dec}(y_1)}^{t=1}(Z)], \\ &\approx \mathbb{E}_{Z \sim q(Z|X)}[\text{Transformer}_{\text{dec}(y_1)}^{t=1}(Z)]. \end{aligned} \quad (68)$$

Similarly:

$$\begin{aligned} \mathbb{E}[y|do(t=0)] &= \int_Z \text{Transformer}_{\text{dec}(y_0)}^{t=0}(Z)p(Z|X)dZ, \\ &= \mathbb{E}_{Z \sim p(Z|X)}[\text{Transformer}_{\text{dec}(y_0)}^{t=0}(Z)], \\ &\approx \mathbb{E}_{Z \sim q(Z|X)}[\text{Transformer}_{\text{dec}(y_0)}^{t=0}(Z)]. \end{aligned} \quad (69)$$

The Individual Treatment Effect (ITE) is:

$$\begin{aligned} \text{ITE}(X) &= \mathbb{E}[y|do(t=1), X] - \mathbb{E}[y|do(t=0), X], \\ &= \mathbb{E}_{q(Z|X)}[\text{Transformer}_{\text{dec}(y_1)}^{t=1}(Z) - \text{Transformer}_{\text{dec}(y_0)}^{t=0}(Z)]. \end{aligned} \quad (70)$$

Final result:

$$\text{ITE}(X) = \mathbb{E}_{q(Z|X)}[\text{Transformer}_{\text{dec}(y_1)}^{t=1}(Z) - \text{Transformer}_{\text{dec}(y_0)}^{t=0}(Z)]. \quad (71)$$

This completes the proof that under the defined causal structure and latent variable model, the *ITE* is identifiable and derivable by integrating over the inferred latent confounders

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Data Availability The Infant Health and Development Program (IHDP) dataset can be downloaded from <https://www.fredjio.com/>. The source code and pretrained models are publicly available at <https://github.com/Statistics112233/TCEVAE>, enabling full reproducibility of our results.

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